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Project Title:	Dynamics and Structure in Colloidal Suspensions Far from Equilibrium		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Element:	None		
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	A dense pack of solid particles can be fluidized by an upward flow of fluid, which counterbalances the gravitational force on the particles. A liquid-fluidized bed of small particles (of the order of 10-100 microns) can be prepared in a uniformly fluidized state; i.e. one without large-scale density fluctuations. In the case of gas-fluidized beds however, numerous studies have found that the particles are unstable to gas bubble formation just at the onset of uniform fluidization. An understanding of the dynamics of gas-fluidized beds raises interesting and important scientific issues: What is the role of inertia? What are the limits of stability? What are the characteristic patterns of the unstable flows? The primary difficulty in studying gas-fluidized beds experimentally is that the buoyancy mismatch makes the gravitational force on the particles unavoidably large. These systems can only be fluidized by high-velocity gas flows where the dynamics is immediately very complicated, and the bed is unstable as soon as it is fluidized. Thus it is not known whether or not gas-fluidized beds are stable to uniform fluidization at sufficiently low Stokes number, as would		

Task Description:	be expected from purely fluid-dynamical considerations, or whether other forces, such as cohesion, play a destabilizing role. These critical questions can not be addressed because stable gas fluidization can not be achieved in earth's gravity. In order to study the onset of a particle-induced instability, the minimum fluidization velocity must be reduced by several orders of magnitude. This can only accomplished in a microgravity environment. In our current NASA sponsored research we have focused on the sedimentation of monodisperse liquid-particle suspensions. Numerical simulations have led to a breakthrough in understanding the origin of a correlation length that appears spontaneously in a settling suspension. Nevertheless laboratory experiments have raised doubts that our understanding of the dynamics at low Reynolds number is complete. In particular, the experiments raise issues of the effects of system size, initial conditions, and polydispersity, which have been largely ignored up to now. Our new proposal will build on these prior accomplishments. Its overall objective is to continue our ongoing efforts to develop a better understanding of gas fluidized systems by extending the current limits of laboratory experiments and numerical simulation. We use experiments and simulations synergistically; thus we not only use laboratory experiments to validate the results of numerical simulations, but we also use simulations to clarify aspects of laboratory experiments, such as polydispersity, that are difficult to control experimentally. Specifically we will attempt to answer three fundamental questions related to particle-fluid flows, motivated in part by what was discovered in the past 3 years research. The first question is to understand the origin of the observed correlation length in settling suspensions and decide if it is universal, applying also to fluidized suspensions. The second issue is to discover if particle collisions in a low-Reynolds number fluidized bed lead to important changes in suspension microstructure at finite Stokes numbers, and to investigate whether such microstructural changes are related to the onset of instability in gas-fluidized suspensions at low Stokes numbers. Finally we want to examine a flow with a non-uniform external force, namely a rotating suspension flow. In this case the centrifugal force produces a non-uniform body force on the particles, leading to a surprising segregation of particles along the rotation axis into bands of high and low concentration. The bands of high and low particle concentration may develop differently in a microgravity environment than on earth. We propose a coordinated program of numerical simulations and laboratory experiments to accomplish the proposal objectives. Numerical simulations will be used to probe parameter regimes that are unavailable to laboratory experiments, and also to assess t [abstract truncated]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will directly impact our knowledge of the stability of many complex fluids that are susceptible to phase separation due to gravity. Understanding the nature of the phase separation will enable us to determine strategies to prevent it. It will help improve the shelf life of many consumer products.
Task Progress:	The project has been completed and several papers are currently in press. A full report is supplied in the Annual report. As a summary: The velocity fluctuations of particles in a low Reynolds number fluidized bed have important similarities and differences with the velocity fluctuations in a low Reynolds number sedimenting suspension. We show that like sedimentation, the velocity fluctuations in a fluidized bed are well described by the balance between density fluctuations due to Poisson statistics and Stokes drag. However, unlike sedimentation, the correlation length of the fluctuations in a fluidized bed increases with volume fraction. We argue that this difference arises because the relaxation time of density fluctuations is completely different in the two systems. To conclude, we have presented the first direct experimental evidence demonstrating that local velocity fluctuations in a fluidized bed are controlled by the balance between locally Poisson density fluctuations and Stokes drag [P. J. Mucha, S.-Y. Tee, D. A. Weitz, B. I. Shraiman, and M. P. Brenner, A model for velocity fluctuations in sedimentation, J. Fluid Mech. 501, 71 (2004).]. However, the local correlation length in a fluidized bed increases with volume fraction in striking contrast to sedimentation. We argue that the difference between these two seemingly similar experiments is caused by the different dynamics of small density fluctuations: in the fluidized bed the dynamics is dictated by the balance between the upflow from the pump and the backflow from the particles. The study of velocity fluctuations in the sedimentation of spheres is complicated by the time evolution of the underlying particle distribution, both at the microscale and in the bulk. We perform a series of experiments and simulations to isolate the effect of an initial, stable stratification in the particle concentration. The directly observed dependence of velocity fluctuations on stratification agrees with a previously-obtained scaling theory. Our initially-stratified experiments and simulations are each consistent with the model for stratification suppression of velocity fluctuations, but with different constants setting the crossover stratifications. Other controlling mechanisms not studied here could of course prevail in other settings. Moreover, it is natural to question whether this scaling even makes sense at large stratifications where the suppressed correlation length becomes less than the interparticle scale. We note that the smallest swirl sizes deduced from l/d here correspond to a couple mean interparticle spacings in the simulations and are even below the mean spacings in the most stratified experiments; meanwhile, we observe that the simulation data appears to indicate greater suppression of fluctuations beyond the model scaling at the largest stratifications considered. It remains possible that the Poisson model needs modification for larger stratifications and cell-geometry effects, though it appears reasonably consistent with the experiments and simulations considered here.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Tee S-Y, Mucha PJ, Brenner MP, Weitz DA. "Velocity fluctuations in a low-Reynolds-number fluidized bed." Journal of Fluid Mechanics, 2008 Jan;596:467-75. http://dx.doi.org/10.1017/S0022112007009652 , Jan-2008
Articles in Peer-reviewed Journals	Tee S-Y, Mucha PJ, Brenner MP, Weitz DA. "Velocity fluctuations of initially-stratified sedimenting spheres." Physics of Fluids, 2007 Nov;19(11):113304. http://dx.doi.org/10.1063/1.2806597 , Nov-2007

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